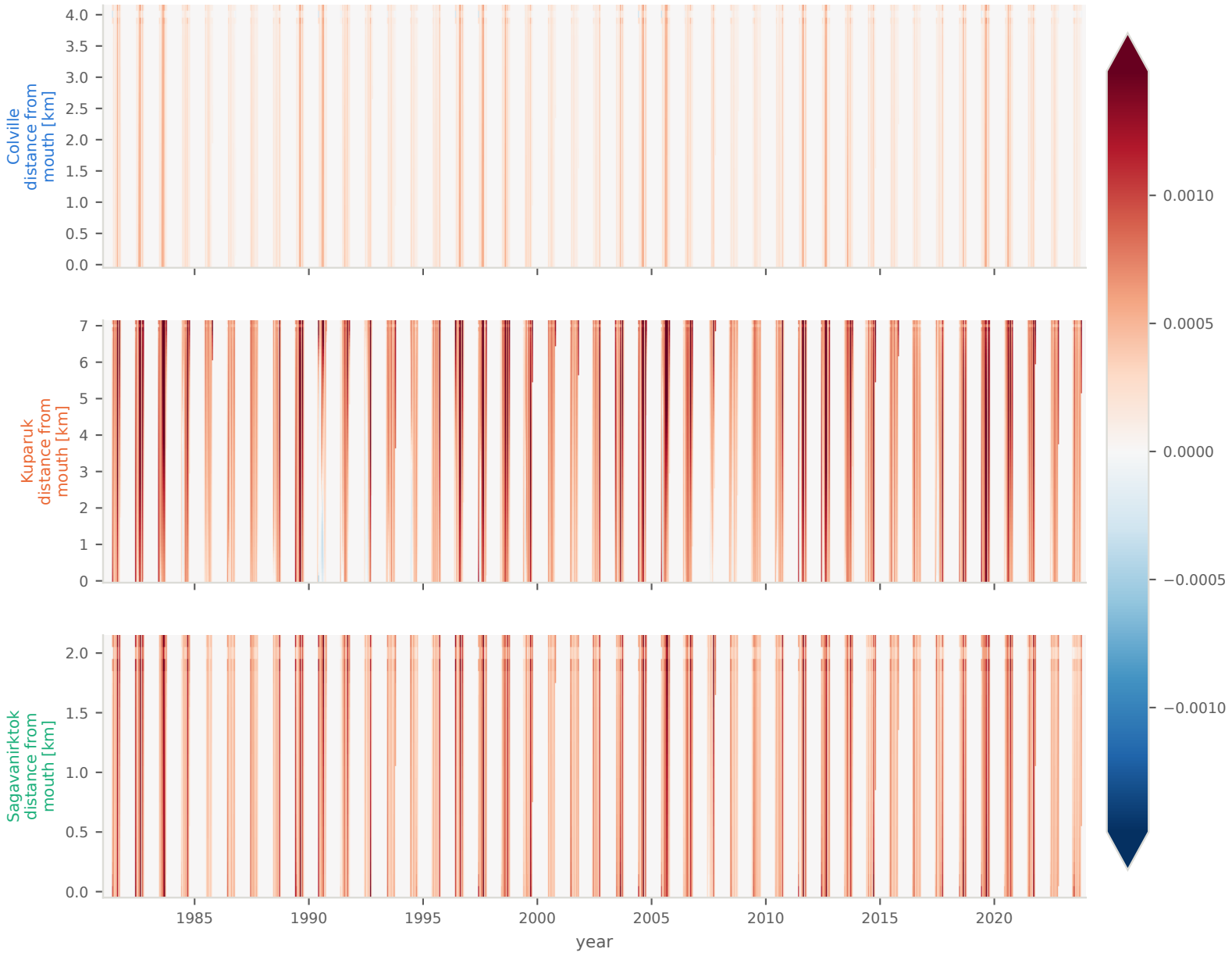
air–sea CO_2 flux [mmol C m^{-3} s^{-1}] (>0 outgassing) — distance × time (1981–2023, daily)



Upstream boundaries (except T, and TOC in the 44-yr record) are CONSTANT -- structure shown here is the model's own internal dynamics acting on a constant source, not a real time-varying river signal. See CLAUDE.md -> "Publication figure suite".